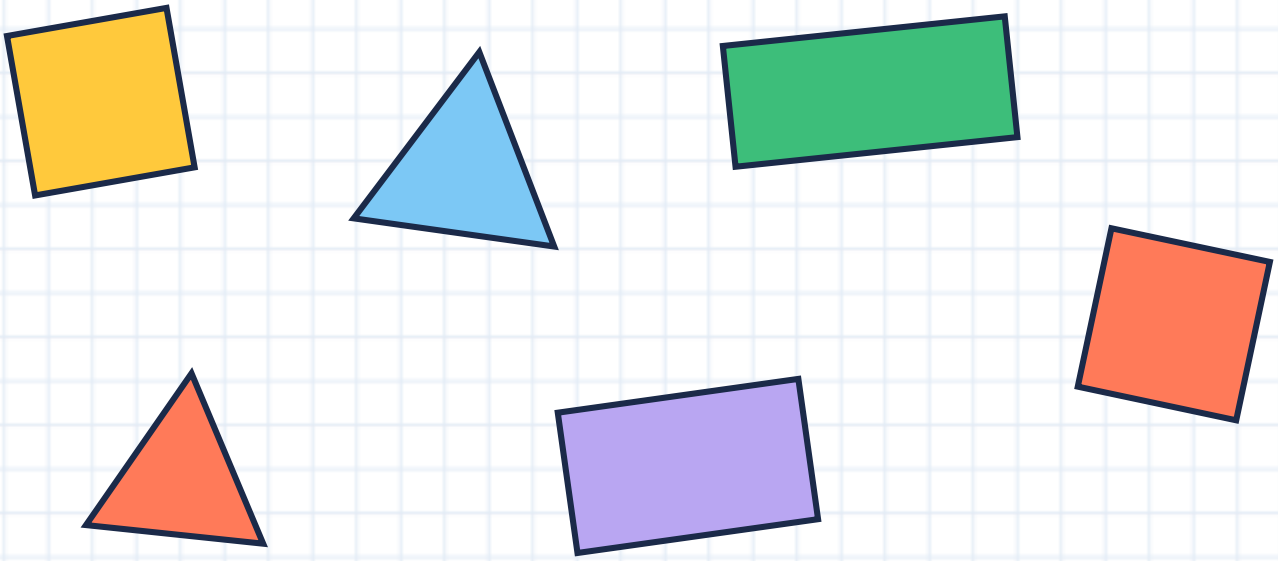


LEVEL 6 · HOLIDAY PROGRAM

Math Inquirers

Notice it. Test it. Explain why.

Name _____ **Group** _____

My Inquirer Rules

1. Notice the pattern.
2. Test it with objects first.
3. Check your answer.
4. Explain why.

My Inquiry Journal

Your tutor stamps each lesson.

Lesson 1
Code the Pattern



Lesson 2
Prime Hunters



Lesson 3
Below Zero



Lesson 4
Algebra Machines



Lesson 5
Every Possible Way



Lesson 6
Fair or Unfair?



Lesson 7
Turtle Geometry



Lesson 8
Secret Codes



Lesson 9
Data Detectives



Lesson 10
Showcase Challenge



Tutor: _____ Program dates: _____

CODE THE PATTERN

Start here Know this idea already? You may start with Think harder. Or after question 3 choose: carry on, or try Think harder.

Name _____ Group _____ Date _____

No.1

Matchstick squares. Each new pattern adds one more square. Fill in the table.



Pattern 1 Pattern 2 Pattern 3

Pattern n	1	2	3	4	5	6
Sticks	4					

a Each new square adds _____ sticks. Why not 4? _____

b Complete the rule: sticks = _____ × n + _____

c Pattern 100 has _____ sticks. (d) Which pattern has 61 sticks? _____

No.2

For each sequence, write the difference, the rule for the nth term and the 100th term. (Two more are on the Think harder page.)

Sequence	Difference	nth term	100th term
(a) 6, 10, 14, 18, ...			
(b) 1, 6, 11, 16, ...			

No.3

Look at the sequence 5, 9, 13, 17, ...

a nth term: _____ (b) Is 101 a term? Which one? _____

c Is 150 in the sequence? Explain. _____

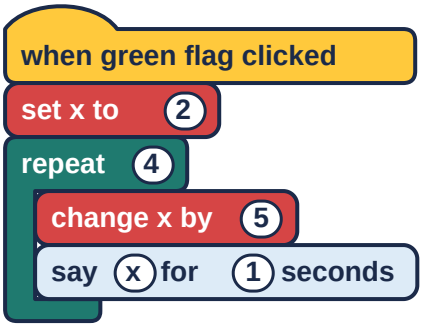
```
term = 5
for pos in range(1, 101):
    print(pos, term)
    term = term + 3
```

Python corner (optional): a term tester

It prints the first 100 terms of 5, 8, 11, ... Change the 5 and the 3 to check your answers to Q1 and Q2.

No.4

Predict, then trace. What does this code make the sprite say? Fill in the trace table.



Loop	x after "change"
1	
2	
3	
4	

(a) The last number it says: _____

b With repeat 100, what is the last number? Explain without running it.

.....

No.5

Plan your code. Use only these blocks to say the first 20 terms of 4, 10, 16, 22, ...

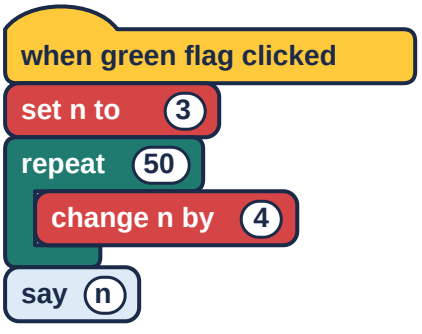


My program:

(a) The last number it says: _____

No.6

Spot the bug. Mei wants her sprite to say the 50th term of 3, 7, 11, 15, ... Here is her code.



(a) Her code says: _____
(b) Use the rule for the nth term.
The 50th term is really: _____

c Fix her code. Explain the bug in one sentence.

.....

FINISH PAGE

Name _____ Group _____ Date _____

Everyone gets to this page. The investigation is for anyone who wants it, at any time. The exit slip and looking back are for everyone.

Investigation **What if we change the shape?**

Triangles in a row use 3, 5, 7, ... sticks (the rule is 2n + 1). What if we join squares in a row, or pentagons, or hexagons, sharing one side each time? Pick a shape. Draw patterns 1, 2 and 3, make a table and find the rule. What changed? What stayed the same?

First step: draw 3 squares in a row and count the sticks.

Exit slip **The sequence 8, 11, 14, 17, ... (a) Write the rule for the nth term. (b) Find the 100th term. (c) A loop starts at the 1st term. How many times must it repeat "change by 3" to reach the 100th term?**

Looking back About 2 minutes. Nobody marks this. You can write "pass" for any line.

Something I did well _____

The hardest part was _____

What helped me: [ ] I drew or built it [ ] I tried a smaller example [ ] I checked my answer [ ] I asked for help

[ ] I am still stuck (my tutor will check in during the lesson) [ ] pass

STORY CORNER **Grace Hopper** (United States, 1906 to 1992)

Grace Hopper was a maths teacher who became a Navy computer scientist. In 1952 she finished an early compiler, a program that turns English-like instructions into code a computer can run. In 1947 her colleagues taped a real moth into a logbook, but engineers had used the word bug long before that.

Try it (optional): A loop starts at 3 and adds 4 each time: 3, 7, 11, 15. Write the nth term, then predict the 10th term without running the loop.

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Choose any question. You can start here after question 3, or when you finish the Start here pages. You can go back to Start here at any time. Everyone joins Talk time and the exit slip.

No.1

Find the 100th term of 12, 19, 26, 33, ...

No.2

Is 200 a term of 2, 9, 16, 23, ...? Explain.

No.3

Matchstick houses in a row: 1 house uses 6 sticks, 2 houses use 11, 3 houses use 16. How many houses can you make with 100 sticks?

No.4

Write the difference, the rule for the nth term and the 100th term of (a) 9, 11, 13, 15, ... and (b) 0.5, 1, 1.5, 2, ...

Go further

No.5

Make up your own sequence puzzle. Choose a first term and a difference, write the first four terms, then ask: "Is ____ a term?" Pick a number that is NOT a term but is close to one. Swap with a partner and explain how you know.

No.6

Which numbers below 60 are in BOTH sequences 2, 5, 8, 11, ... and 4, 9, 14, 19, ...? Find them all.

No.7

What does this code say? set t to 0, set k to 0, then repeat 5 [change k by 1, change t by k], then say t. What would it say with repeat 100?

PRIME HUNTERS

Start here Know this idea already? You may start with Think harder. Or after question 2 choose: carry on, or try Think harder.

Name _____ Group _____ Date _____

No.1

A prime number has exactly 2 factors: 1 and itself. Use factor pairs to fill in the table.

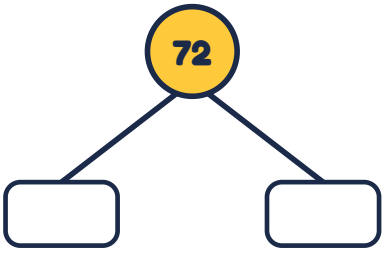
Number	All its factors	How many?	Prime?
28			
31			
49			
51			
1			

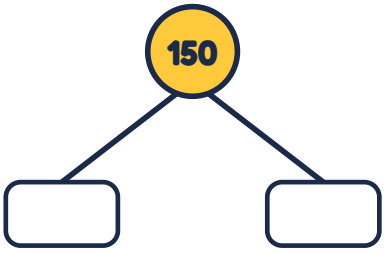
a Test with 2, 3, 5 and 7. Circle the primes: 57 67 77 83 97 99

b Ali says 87 is prime because it is odd and does not end in 5. Is he right?

No.2

Factor trees. Keep splitting until every branch ends in a prime.





a 72 = _____ (b) 150 = _____

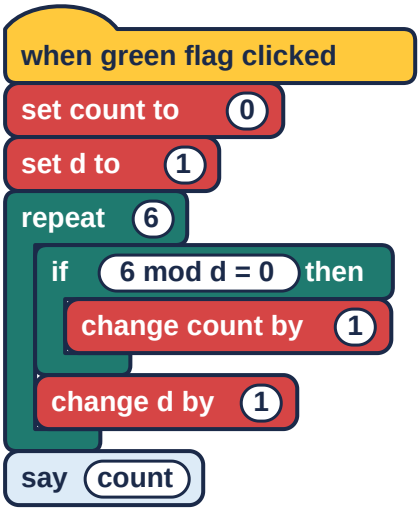
c A number is 2 × 3 × 3 × 7. What is the number? _____

Python corner (optional): a factor finder. n % d is n mod d. It lists the factors of n and counts them. Change n to check Q1.

```
n = 28
f = [d for d in range(1, n + 1) if n % d == 0]
print(f, len(f))
```

No.3

Predict, then trace. This code counts the factors of 6. Fill in the trace table.



d	6 mod d	count after
1	0	1
2		
3		
4		
5		
6		

(a) What does it say at the end? _____

(b) Change both 6s to 13. What does it say? Why?

.....

No.4

Goldbach's guess: every even number above 2 is a sum of two primes.

a 24 in every way: _____

b Explain why 27 cannot be the sum of two primes.

No.5

Exactly 3 factors.

a All the numbers from 1 to 30 with exactly 3 factors (there are 3):

_____

b What do you notice? _____

No.6

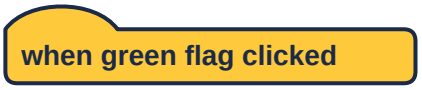
Plan your code. Put the blocks in order so the sprite says every factor of 40.

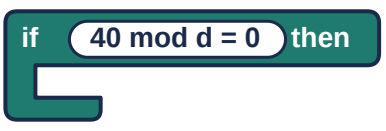
A 

B 

C 

D 

E 

F 

My order (show what goes inside):

.....

How many numbers does it say? _____

FINISH PAGE

Name _____ Group _____ Date _____

Everyone gets to this page. The investigation is for anyone who wants it, at any time. The exit slip and looking back are for everyone.

Investigation **Is it always true?**

Ali says: "If you add two different primes, the answer is never a prime." Test him. List every pair of different primes below 20 and add them. Is Ali right? What do you notice about the pairs that do make a prime? Explain why.

First step: write down all the prime numbers below 20.

Exit slip **(a) Is 111 prime? Show your test. (b) Write 66 as a product of primes. (c) Optional: is 133 prime?**

Looking back About 2 minutes. Nobody marks this. You can write "pass" for any line.

Something I did well .....

The hardest part was .....

What helped me: [ ] I drew or built it [ ] I tried a smaller example [ ] I checked my answer

[ ] I asked for help

[ ] I am still stuck (my tutor will check in during the lesson) [ ] pass

STORY CORNER **Kamal al-Din al-Farisi** (Persia, about 1260 to about 1320)

Kamal al-Din al-Farisi lived in Persia about 700 years ago. He studied rainbows by shining light through a water-filled glass sphere in a dark room. He also used prime factors to check numbers and worked with amicable pairs such as 220 and 284.

Try it (optional): Write 284 as a product of primes. List its factors smaller than 284 and add them. What do you get?

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Choose any question. You can start here after question 2, or when you finish the Start here pages. You can go back to Start here at any time. Everyone joins Talk time and the exit slip.

No.1

How many factors does 64 have? List them.

No.2

Find the smallest number that has exactly 6 factors.

No.3

Write 100 as the sum of two primes in every possible way.

No.4

Write 36 as the sum of two primes in every way. Then find ALL the numbers from 1 to 50 with exactly 3 factors. What is the next one after 50?

Go further

No.5

Make up a “Who am I?” number puzzle using factors and primes, for example: “I am between 30 and 60 and I have exactly three factors.” Write 2 or 3 clues so that only ONE number fits. Swap with a partner, then check that no other number fits.

No.6

A farmer counts her eggs in 2s, 3s, 4s, 5s and 6s, and each time there is 1 egg left over. When she counts them in 7s, none are left over. What is the smallest number of eggs she could have?

No.7

20 lockers are closed. Student 1 opens every locker. Student 2 changes every 2nd locker (open to closed, closed to open). Student 3 changes every 3rd locker, and so on up to student 20. Which lockers are open at the end?

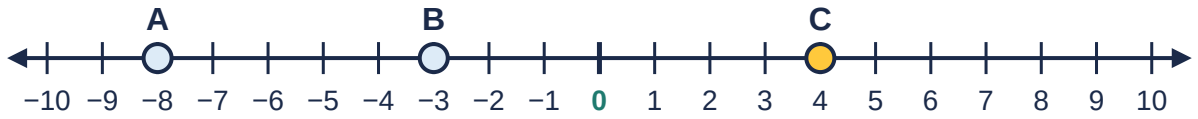
BELOW ZERO

Start here Know this idea already? You may start with Think harder. Or after question 3 choose: carry on, or try Think harder.

Name _____ Group _____ Date _____

No.1

The number line. Numbers get bigger to the right.



a Write the number at each letter. A = _____ B = _____ C = _____

b Smallest first: 3, −5, 0, −1, −12, 7 → _____

c Write < or > in each gap. −4 _____ −9 −1 _____ 0 −15 _____ 2

No.2

On a December trip to Harbin, China, Mei wrote down the temperature.

Time	6 a.m.	9 a.m.	12 noon	3 p.m.	9 p.m.
Temperature	−24 °C	−19 °C	−13 °C	−11 °C	−20 °C

a At which time was it warmest? _____

b How many degrees did it rise from 6 a.m. to 12 noon? _____

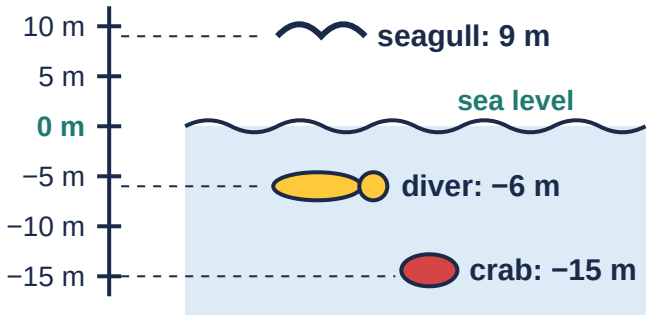
c How many degrees did it fall from 3 p.m. to 9 p.m.? _____

d Midnight was 5 degrees colder than 9 p.m. What was the temperature? _____

e At noon Singapore was 31 °C. How much warmer was it than Harbin? _____

No.3

Heights above sea level are positive. Heights below sea level are negative.



(a) How far above the diver is the seagull? _____

(b) How far below the diver is the crab? _____

(c) The diver swims up 4 m, then down 11 m. Her height now? _____

No.4

Work these out. Draw the jumps on the number line to help you.

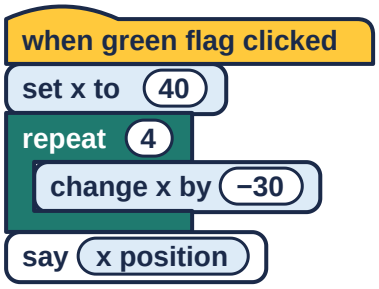


a $-6 + 9 =$ _____ (b) $4 - 11 =$ _____ (c) $-3 - 5 =$ _____

d $-2 + (-6) =$ _____ (e) $5 - (-3) =$ _____ (f) $-1 - (-7) =$ _____

No.5

Trace the code. Fill in the table to show x after each repeat.



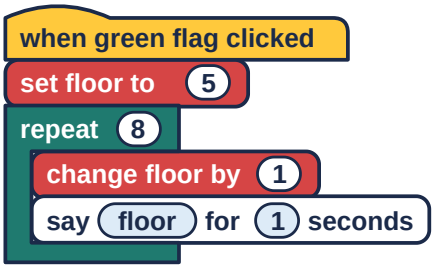
repeat	start	1	2	3	4
x	40				

(a) What does it say at the end? .....

(b) How many MORE repeats would take it to x = −200? .....

No.6

Find the bug. Siti wants the lift to go DOWN 8 floors from floor 5. (0 = ground floor)



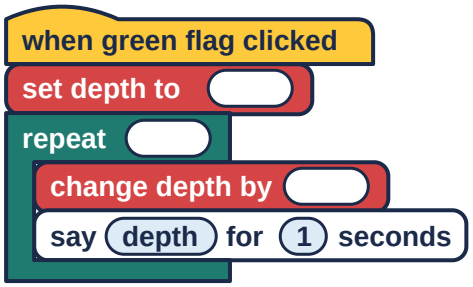
(a) Her code says this floor last: .....

(b) Change ONE block so the lift goes down. New block: .....

(c) Now the last floor it says is which basement? .....

No.7

Plan your code. A diver starts at sea level (0 m) and goes down 4 m at a time to −28 m.



(a) Fill in the three blanks.

(b) After the loop she is at −28 m.
One more block: change depth by _____
What number brings her back to 0 m?

Know Python? Today's thermometer:

```
temp = 5
for hour in range(6):
    temp = temp + (-3)
    print(temp)
```

FINISH PAGE

Name _____ Group _____ Date _____

Everyone gets to this page. The investigation is for anyone who wants it, at any time. The exit slip and looking back are for everyone.

Investigation **Find all the pairs**

Here are some numbers: –5, –3, –1, 0, 2, 4, 6. Find ALL the pairs (a, b) from the list with a – b = 7. Mark each pair on a number line. What do you notice? Then try again with a – b = 3.

First step: draw a number line from –5 to 6. Pick a = 2 and ask: what number b makes 2 – b = 7?

Exit slip **It is –3 °C. The temperature falls 6 degrees, then rises 4 degrees. What is the temperature now?**

Looking back About 2 minutes. Nobody marks this. You can write "pass" for any line.

Something I did well _____

The hardest part was _____

What helped me: [ ] I drew or built it [ ] I tried a smaller example [ ] I checked my answer [ ] I asked for help

[ ] I am still stuck (my tutor will check in during the lesson) [ ] pass

STORY CORNER **Brahmagupta** (India, 598 to about 670)

Brahmagupta wrote rules for zero and negative numbers in the year 628. He called positive numbers fortunes and negative numbers debts. Not every rule he wrote was right, and that is part of how maths grows: people check each other's ideas.

Try it (optional): Debts and fortunes: work out (-6) × (-4). Then work out (-8) - (-3).

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Choose any question. You can start here after question 3, or when you finish the Start here pages. You can go back to Start here at any time. Everyone joins Talk time and the exit slip.

No.1

A sprite starts at x = a. It runs "change x by –8" and then "change x by 5". It ends at x = –1. What was a? Work backwards from the end, and check by running it forwards.

No.2

A diver is at –14 m. A seagull is flying at 9 m. How far apart are they?

No.3

Use the numbers –2, 3 and 5 once each to fill the gaps in _____ – _____ – _____. What is the biggest answer you can make? What is the smallest?

No.4

City A is 5 degrees colder than city B. City B is 3 degrees warmer than city C. City C is at –4 °C. Find the temperature of B and of A.

Go further

No.5

Make up your own temperature story with at least three changes (rises and falls) that starts below zero and ends at exactly –7 °C. Write the story, the working and a number-line sketch. Swap with a partner and check each other's.

No.6

Two numbers add up to –2. Their difference is 10. What are the two numbers?

No.7

In some Singapore buildings there is no level 0. (Q6 on page 2 uses 0 for the ground floor; here, level 1 is the ground floor and B1 is the level below it.) A lift goes from level 4 down to B3. How many floors does it go down? (Count the levels one by one; do not use a number line with 0.)

ALGEBRA MACHINES

Start here Know this idea already? You may start with Think harder. Or after question 3 choose: carry on, or try Think harder.

Name _____ Group _____ Date _____

No.1

Guess my rule. Fill in the gaps. Write each rule with n.

(a) IN 1, 2, 3, 4 give OUT 4, 5, 6, 7. Rule: OUT = n + ____ OUT for IN 10 = ____

(b)

IN	1	2	3	4	10
OUT	6	9	12	15	

Rule: OUT = ...

(c)

IN	1	2	3	4	
OUT	3	8	13	18	48

Rule: OUT = ...

(d)

IN	1	2	3	4	20
OUT	1	5	9	13	

Rule: OUT = ...

No.2

Two machines use the same steps in a different order.

Machine P: n → + 3 → × 4 → OUT

Machine Q: n → × 4 → + 3 → OUT

a Put in 5. Machine P gives _____ Machine Q gives _____

b Write each rule with n. P: OUT = _____ Q: OUT = _____

c Can P and Q ever give the same OUT for the same IN?
Explain. _____

No.3

Run each machine backwards to find n. Then check by going forwards.

a n → × 6 → – 5 → 43 n = _____ Check: _____

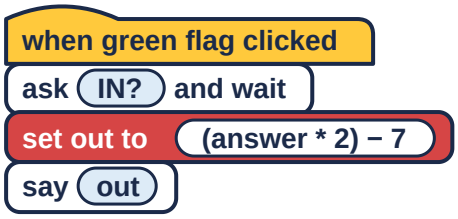
b 7n + 4 = 74 n = _____ Check: _____

c (n – 3) × 5 = 45 n = _____ Check: _____

d n ÷ 4 + 9 = 15 n = _____ Check: _____

No.4

Predict what the code says. (In Scratch, * means ×.)



(a) You type 10. It says _____

(b) You type 3. It says _____

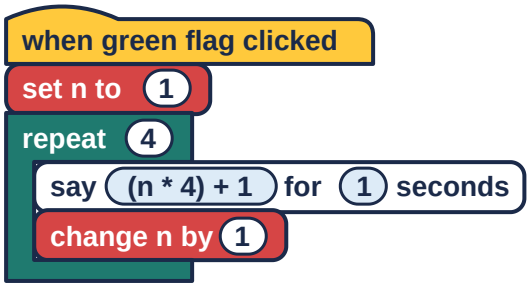
(c) To make it say 25, type _____

Know Python? The explore machine:

```
n = int(input("IN? "))
out = n * 3 + 2
print("OUT:", out)
```

No.5

Trace the code. Fill in the table.



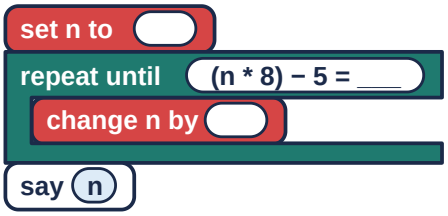
repeat	n	it says
1	1	
2		
3		
4		

(b) The rule: it says _____

(c) With repeat 10, the last number is _____

No.6

Plan a search. Find n when 8n – 5 = 91 by trying n = 0, 1, 2, 3 ...



(a) Fill in the three blanks.

(b) What will it say? _____

(c) Check by running the machine backwards: _____

No.7

The number trick. Think of a number. Add 5. Double it. Take away 4. Halve it. Take away the number you first thought of.

a Try it with 7: you get _____. Try it with 10: you get _____.

b Use n: n → n + 5 → 2n + 10 → _____ → _____ → _____ (the first steps are done for you)

c Explain why you always get the same answer.

_____

FINISH PAGE

Name _____ Group _____ Date _____

Everyone gets to this page. The investigation is for anyone who wants it, at any time. The exit slip and looking back are for everyone.

Investigation **Find all the machines**

A machine does "× a number, then + a number". Use whole numbers: the × number is from 1 to 9 and the + number is from 0 to 9. Find ALL the machines that turn IN 4 into OUT 17. What pattern do you spot in your answers? Then try IN 4 and OUT 21.

First step: try × 2. Then 4 × 2 = 8. What must be added to 8 to make 17?

Exit slip **A machine does × 4, then + 5. (a) IN 6: what is OUT? (b) OUT 45: what was IN?**

Looking back About 2 minutes. Nobody marks this. You can write "pass" for any line.

Something I did well _____

The hardest part was _____

What helped me: [ ] I drew or built it [ ] I tried a smaller example [ ] I checked my answer [ ] I asked for help

[ ] I am still stuck (my tutor will check in during the lesson) [ ] pass

STORY CORNER **Al-Khwarizmi** (Khwarazm, about 780 to about 850)

Al-Khwarizmi worked in Baghdad about 1,200 years ago. His book on solving equations gives us the word algebra, from al-jabr, and his name gives us the word algorithm. He wrote everything in words, with no letters such as x.

Try it (optional): A machine multiplies the input by 3, then adds 2. The output is 20. Balance 3x + 2 = 20 to find x.

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Choose any question. You can start here after question 3, or when you finish the Start here pages. You can go back to Start here at any time. Everyone joins Talk time and the exit slip.

No.1

IN 1, 2, 3, 4 gives OUT 1, 7, 13, 19. Find the rule. What is OUT when IN is 20?

No.2

Solve $9n - 7 = 65$.

No.3

Machine A does × 2, then + 10. Machine B does + 5, then × 2. Put 7 into both. What do you notice? Explain with n.

No.4

I think of a number, divide it by 4, then add 6. I get 15. What is my number?

Go further

No.5

Find two different machines of the form "× a number, then + or – a number" that both turn IN 5 into OUT 18. Then find the machine of this form that ALSO turns IN 2 into OUT 6. Is there only one?

No.6

Mei is n years old. Her mother is 3 times as old. In 12 years, her mother will be twice as old as Mei. How old is Mei now?

No.7

Ali's secret machine does × a number, then + or – a number. IN 4 gives OUT 9. IN 9 gives OUT 24. What does IN 20 give? Which IN gives OUT 99?

EVERY POSSIBLE WAY

Start here Know this idea already? You may start with Think harder. Or after question 3 choose: carry on, or try Think harder.

Name _____ Group _____ Date _____

No.1

Mei packs 4 T-shirts, 3 pairs of shorts and 2 caps for camp.

a Outfits of 1 T-shirt and 1 pair of shorts: how many? _____ (b) With a cap too? _____

No.2

Digit cards. Use each card at most once in a number. A number cannot start with 0.

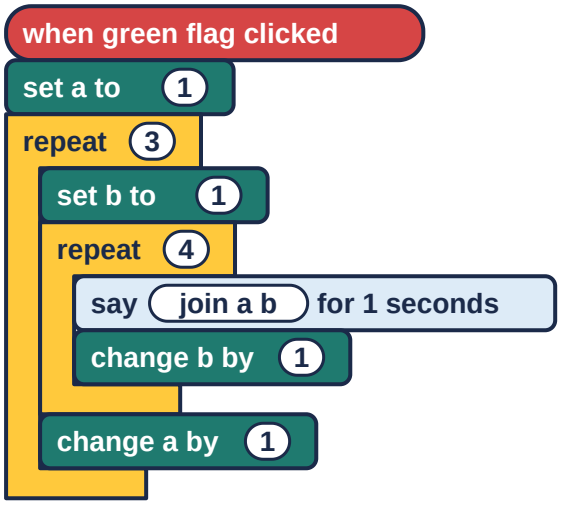
0	3	5	8

(a) How many 2-digit numbers can you make? .....
List them from smallest to largest:
.....

(b) How many 3-digit numbers? Write the multiplication.
.....

No.3

Trace the code. (a) Fill in a, b and what the sprite says, for the first 6 things it says.



a	b	Sprite says

(b) How many things does it say altogether? .....

(c) The last thing it says? .....

(d) If "repeat 3" becomes "repeat 5", how many things will it say? .....

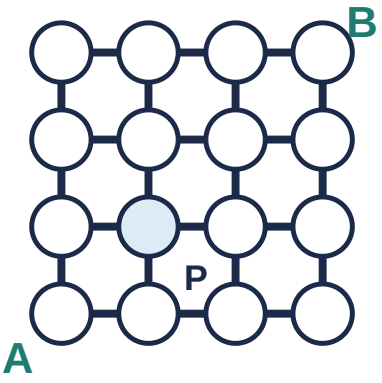
No.4

Seven friends meet. Every pair of friends shakes hands once. (a) How many handshakes are there? Show your method.

b Ali says, "7 × 6 = 42 handshakes." What is wrong? Fix it.

No.5

Routes on a grid. Walk along the lines from A to B, going only right or up.



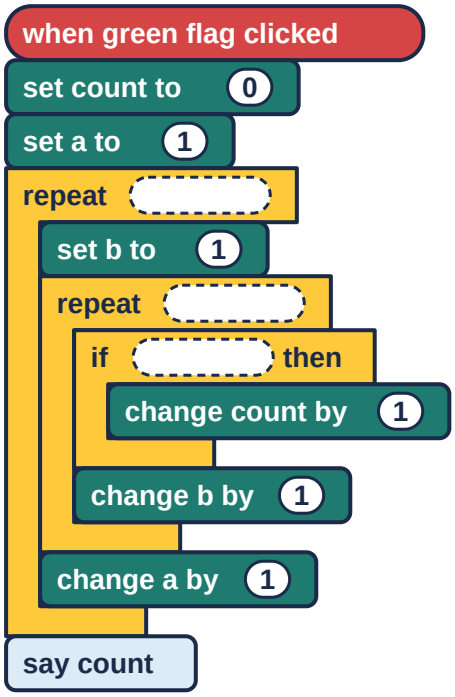
Write in each circle the number of routes from A to it. A gets 1. Each other circle = the number on its left + the number below it.

(a) How many routes go from A to B? .....

(b) A puddle covers the blue point P. Do it again with 0 at P. How many routes avoid P? .....

No.6

Plan your code. Fill the 3 gaps so it counts the handshakes of the 7 friends in Q4.



(a) What will the program say? .....

Does it agree with Q4(a)? .....

(b) If the test is changed to **a = b**, what will it say? What is it counting now?

.....

Python corner (optional): our Explore program. Change the lists to test Q1.

```
teas = ["milk", "green", "taro"]
toppings = ["pearls", "jelly"]
count = 0
for tea in teas:
    for top in toppings:
        print(tea, "tea with", top)
        count = count + 1
print(count, "orders")
```

FINISH PAGE

Name _____ Group _____ Date _____

Everyone gets to this page. The investigation is for anyone who wants it, at any time. The exit slip and looking back are for everyone.

Investigation **Find all the ways**

Pay exactly 50 cents using only 5-cent, 10-cent and 20-cent coins (as many of each as you like). Find ALL the ways. Show how you made sure you did not miss one or count one twice. What if there is also a 50-cent coin?

First step: list every way to pay 50 cents that uses two 20-cent coins.

Exit slip **Six friends meet. Every pair of friends shakes hands once. How many handshakes are there? Show how you know.**

Looking back About 2 minutes. Nobody marks this. You can write "pass" for any line.

Something I did well _____

The hardest part was _____

What helped me: [ ] I drew or built it [ ] I tried a smaller example [ ] I checked my answer

[ ] I asked for help

[ ] I am still stuck (my tutor will check in during the lesson) [ ] pass

STORY CORNER **Yang Hui** (China, about 1238 to about 1298)

Yang Hui lived in China more than 750 years ago. In 1261 he wrote down a triangle of numbers where each inside number is the sum of the two above it, and he credited an earlier mathematician, Jia Xian. In Europe it is called Pascal's triangle, but Pascal wrote about it nearly 400 years later.

Try it (optional): Write the row after 1, 4, 6, 4, 1. Its third number is the number of ways to pick 2 students from 5. What is it?

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Choose any question. You can start here after question 3, or when you finish the Start here pages. You can go back to Start here at any time. Everyone joins Talk time and the exit slip.

No.1

A set lunch is 1 main, 1 drink and 1 dessert. There are 4 mains and 3 drinks. There are 60 different set lunches. How many desserts are there?

No.2

How many 3-digit numbers can you make from the cards 1, 2, 3 and 4, using each card at most once? How many of them are even?

No.3

In a league of 6 teams, every team plays every other team twice (once at home, once away). How many matches are played?

No.4

(a) Mei packs 4 T-shirts, 3 pairs of shorts and 2 caps. She wants at least 40 outfits with a cap. What is the fewest number of extra pairs of shorts she must buy? (b) A party has 45 handshakes. How many people are there?

Go further

No.5

Make up your own “how many ways” puzzle whose answer is exactly 24. Write the puzzle, show how you counted, and check with a list or a tree. Swap with a partner.

No.6

A set of dominoes has one domino for every pair of spot numbers from 0 to 6, including doubles such as 3 | 3. How many dominoes are in the set?

No.7

How many 3-digit numbers have digits that add up to 5? (For example, 203.)

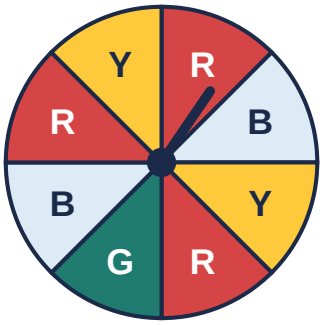
FAIR OR UNFAIR?

Start here Know this idea already? You may start with Think harder. Or after question 3 choose: carry on, or try Think harder.

Name _____ Group _____ Date _____

No.1

Ben's spinner has 8 equal parts: 3 red (R), 2 blue (B), 2 yellow (Y) and 1 green (G). Write each probability as a fraction in its simplest form.



(a) P(red) = ... **(b)** P(not yellow) = ...

(c) Ben makes the spinner in code and spins it 1000 times. About how many reds do you expect? .....

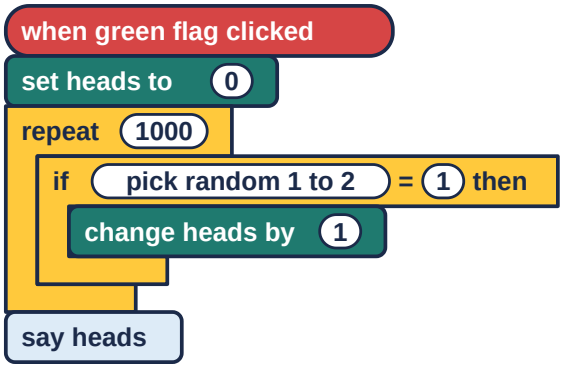
(d) His program says 362 reds. Ben says, "My program is wrong!" Do you agree? Explain. (Hint: a random result is close to the expected number, not exactly on it.)

.....

.....

No.2

Predict what this code does. "pick random 1 to 2" gives 1 or 2, each with probability 1/2.



(a) About what number will it say? .....

(b) Mei runs it 3 times and gets 513, 489 and 502. Is her code broken? Explain.

.....

(c) If "pick random 1 to 2" is changed to "pick random 1 to 5", about what will it say?

.....

No.3

A bag has 4 red and 6 blue counters. You pick one without looking.

a What is the probability of picking red? _____

b How many red counters must be added so that P(red) = 1/2? _____

No.4

The Difference Game. Roll two dice. The difference is the bigger number minus the smaller (0 if they are the same). Ali wins on 0, 1 or 2. Ben wins on 3, 4 or 5.

Difference	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3			0			
4				0		
5					0	
6						0

Python corner (optional): the Explore game. Change it to test Q4.

```
import random
a_wins = 0
b_wins = 0
for game in range(1000):
    d1 = random.randint(1, 6)
    d2 = random.randint(1, 6)
    if 5 <= d1 + d2 <= 9:
        b_wins = b_wins + 1
    else:
        a_wins = a_wins + 1
print(a_wins, b_wins)
```

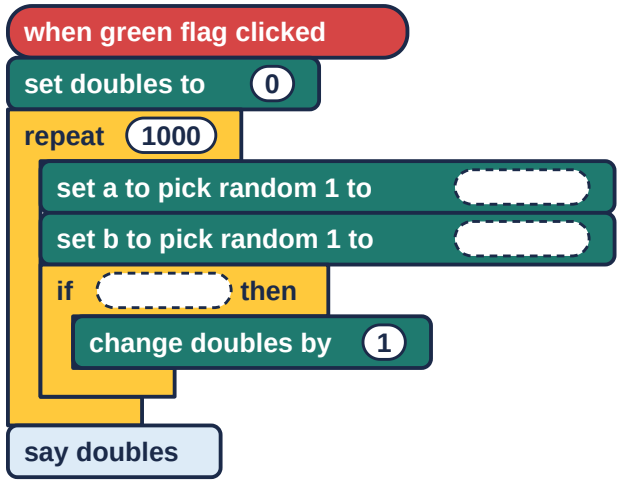
a Complete the table. (b) P(Ali wins) = _____ P(Ben wins) = _____

c Is the game fair? Explain.

_____

No.5

Plan your code. Fill the 3 gaps to count doubles in 1000 rolls of two dice.



(a) About what number will it say? Show your fraction.

.....

(b) Why will it not say exactly 1000 × 1/6?

.....

.....

FINISH PAGE

Name _____ Group _____ Date _____

Everyone gets to this page. The investigation is for anyone who wants it, at any time. The exit slip and looking back are for everyone.

Investigation **Is it always true?**

Ali says: "A game with only two outcomes is always fair." Test him on three games. (1) Flip a coin: heads or tails. (2) Roll a dice: six or not six. (3) Roll two dice: total 7 or not 7. Which games are fair? Is Ali right? Change one unfair game to make it fair.

First step: flip a coin 10 times and tally heads and tails. Is a coin fair? Say why.

Exit slip **Roll two dice and add. What is the probability that the total is 8? If you roll them 360 times, about how many 8s do you expect?**

Looking back About 2 minutes. Nobody marks this. You can write "pass" for any line.

Something I did well _____

The hardest part was _____

What helped me: [ ] I drew or built it [ ] I tried a smaller example [ ] I checked my answer

[ ] I asked for help

[ ] I am still stuck (my tutor will check in during the lesson) [ ] pass

STORY CORNER **Blaise Pascal** (France, 1623 to 1662)

Blaise Pascal built a mechanical calculator to help his father, who worked with taxes. In 1654 he and Pierre de Fermat wrote letters about how to share a prize fairly when a game stops early. Their letters are seen as laying the foundations of probability.

Try it (optional): First to win 3 rounds takes the prize. A has 2 wins, B has 1. List the next two rounds (AA, AB, BA, BB). What fraction of the prize should A get?

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Choose any question. You can start here after question 3, or when you finish the Start here pages. You can go back to Start here at any time. Everyone joins Talk time and the exit slip.

No.1

A spinner has 10 equal parts numbered 1 to 10. What is the probability of spinning a prime number?

No.2

Roll two dice and add. What is the probability that the total is 9?

No.3

A bag has 5 red counters and some blue counters. The probability of picking red is 1/4. How many blue counters are there?

No.4

(a) A bag has 4 red and 6 blue counters. How many blue counters must be added so that P(red) = 1/5? (b) In the Difference Game (Ali wins on 0, 1, 2; Ben on 3, 4, 5), change who wins on which differences so that the game is fair.

Go further

No.5

Design a FAIR game for two players using one dice, two coins or a spinner. Write the rules and show with a list or table that each player has the same chance of winning. Then change ONE rule to make the game unfair, and say who is favoured.

No.6

Mei and Ali each roll one dice. Mei wins if her number is bigger. Ali wins if his number is bigger or the same. Is the game fair? What is the probability that Mei wins?

No.7

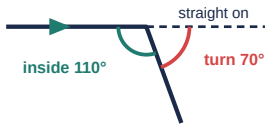
Spinner A has 3 equal parts: 1, 2, 3. Spinner B has 4 equal parts: 1, 2, 3, 4. Spin both and multiply. Even wins if the answer is even, Odd wins if it is odd. Is the game fair?

TURTLE GEOMETRY

Start here Know this idea already? You may start with Think harder. Or after question 2 choose: carry on, or try Think harder.

Name _____ Group _____ Date _____

turn = the swing at a corner.
inside angle = the angle inside the shape (turn + inside = 180°).
total turning = all the turns added up.



No.1

Regular polygons: fill in the table (row 2 is your hexagon from the explore task). Test two rows in Scratch: move 80 up to 10 sides, 50 for 12 to 15, 30 for 24.

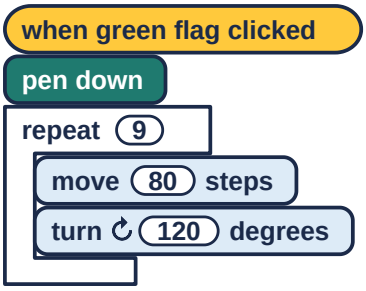
Sides	Turn angle	Inside angle	Sides × turn
3	120°	60°	360°
6			
9			
10			
	24°		
		165°	

a What do you notice about the last column? Why does this happen?

b Mei says a regular polygon can have inside angles of 130°. Is she right? Explain.

No.2

Ben wants a regular nonagon (9 sides). Predict what his program draws, then run it (after the set-up).



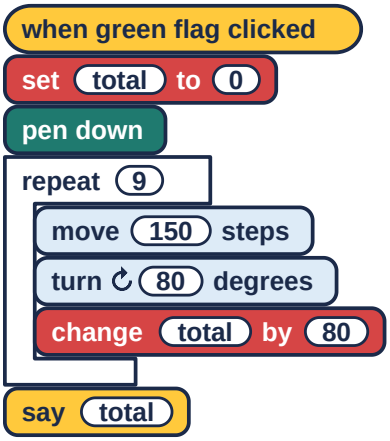
Draw your prediction here

a What does the turtle draw? Explain.

b Total turning _____ ° (c) Fix it: change _____ to _____

No.3

Trace the code. The total is how far the turtle has turned so far. (3 rows are done for you.)



after repeat	total
1	80
2	160
3	240
4	
5	
6	
7	
8	
9	

a The sprite says _____ (b) Times the turtle goes round _____

c It draws a _____ (d) Point angle _____ °

No.4

Plan a star, then test it (move 120 for the 9-point star). Fill in the 9-point row. Turn = 360° × times round ÷ points. Point angle = 180° – turn.

Points	Times round	Total turning	Turn angle	Point angle
5	2	720°	144°	36°
9	4			

No.5

Add up the inside angles of regular polygons (use Q1 for 10 sides).

Sides	3	4	5	6	10
Each inside angle					
All inside angles added					

a One more side adds _____ ° (b) For n sides: total = _____

Optional, for children who already code: Python version

Prints the turn and the total (144.0 720.0). Needs Python turtle, e.g. Trinket.

```
import turtle
t = turtle.Turtle()
n, k = 5, 2   # points, rounds
turn = 360 * k / n
```

```
for i in range(n):
    t.forward(150)
    t.right(turn)
print(turn, n * turn)
turtle.done()
```

FINISH PAGE

Name _____ Group _____ Date _____

Everyone gets to this page. The investigation is for anyone who wants it, at any time. The exit slip and looking back are for everyone.

Investigation **What if the turn changes?**

A turtle repeats [move 100, turn ↻ t] until it first faces the way it started. Make a table for t = 40, 72, 90, 100, 120, 144 and 150: how many repeats, and how many times does it go round? Add up the turns to check. What changes when t changes? What stays the same?

First step: a turtle repeats [move 100, turn 90]. How many repeats until it faces the way it started?

Exit slip **A turtle draws a regular polygon and turns 5° at every corner. How many sides does the polygon have? What is each inside angle?**

Looking back About 2 minutes. Nobody marks this. You can write "pass" for any line.

Something I did well _____

The hardest part was _____

What helped me: [ ] I drew or built it [ ] I tried a smaller example [ ] I checked my answer [ ] I asked for help

[ ] I am still stuck (my tutor will check in during the lesson) [ ] pass

STORY CORNER **Cynthia Solomon** (United States, born 1938)

Cynthia Solomon helped design Logo in 1967, described as the first programming language made for children. Logo's turtle moves relative to itself, so LEFT 90 makes it turn a quarter turn. She taught programming with Logo at a junior high school in 1968 and 1969.

Try it (optional): A turtle draws a regular octagon by repeating FORWARD and RIGHT 8 times. How many degrees does it turn each time? What is one inside angle?

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Choose any question. You can start here after question 2, or when you finish the Start here pages. You can go back to Start here at any time. Everyone joins Talk time and the exit slip.

No.1

Each inside angle of a regular polygon is 4 times its turn angle. How many sides does the polygon have?

No.2

Each inside angle of a regular polygon is 140°. What do all its inside angles add up to?

No.3

Mei's turtle draws a shape with 4 sides. Its first three turns are 70°, 100° and 90°. It comes home facing the same way after going round once. What is its last turn? What is the inside angle at that corner?

No.4

Plan a 20-point star that goes round 3 times (move 100). Find the total turning, the turn angle and the point angle. Which has sharper points, this star or the 9-point star in Q4?

Go further

No.5

Design your own star. Choose the number of points p and how many times it goes round, k, so that the turn angle 360° × k ÷ p is a WHOLE number and p and k share no common factor. Give the turn angle, the point angle (180° – turn) and the code. Explain why your star closes after p repeats.

No.6

A square, a hexagon and a 12-sided shape fit exactly around a point: 90° + 120° + 150° = 360°. Find one other set of three regular polygons whose inside angles add up to 360° (hint: the answer uses a square or a triangle).

No.7

Try turn 120 and turn 60 with repeat 6 and move 100. Which one draws a hexagon? Why does the other not make a 6-point star?

SECRET CODES

Start here Know this idea already? You may start with Think harder. Or after question 2 choose: carry on, or try Think harder.

Name _____ **Group** _____ **Date** _____

Letter numbers (we start at A = 0). Use them for every code question in this lesson.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

No.1

Clock and calendar puzzles. A clock wraps round after 12; a week wraps round after 7 days.

a It is 8 o'clock. 7 hours later it is _____ o'clock. (b) It is 11 o'clock. 30 hours later (count 12 hours at a time): _____ o'clock.

c It is 4 o'clock. 9 hours ago it was _____ o'clock. (d) Today is Saturday. In 100 days: _____ .

e Today is Saturday. 30 days ago it was _____ .

No.2

Caesar codes. Code number = (letter number + shift) mod 26. Test your answers in Scratch (type the message in capital letters; the starter keeps the spaces).

a Code the word PIZZA with a shift of 7. _____

b Decode this message. It was made with a shift of 10: MYNO SC PEX

_____

c Ben wants to decode shift 10 by shifting FORWARD instead. Which shift should he use?

_____

Optional, for children who already code: Python version. % is mod. Change the message and shift to test Q2; a negative shift decodes.

```
alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
message = "MEET AT THE MRT"
shift = 3
code = ""
for letter in message:
    if letter in alphabet:
        n = alphabet.index(letter)   # A is 0
        m = (n + shift) % 26         # % is mod
        code = code + alphabet[m]
    else:
        code = code + letter         # keep spaces
print(code)
```

No.3

Trace the code. The Scratch code machine has a list called alphabet (item 1 is A). For each letter of the message WAY, with shift = 5, it does these three blocks:

```
set [n] to [(item # of (letter) in alphabet) − 1]
set [m] to [(n + shift) mod 26]
set [code] to [join (code) (item (m + 1) of alphabet)]
```

Letter	n	n + 5	m = (n + 5) mod 26	Code letter
W				
A				
Y				

a At the end the program says the code. What does it say? _____

No.4

Crack the code. Mei found this message. It was made with a Caesar shift, but she does not know which one.

BPM SMG QA CVLMZ BPM UIB

a Which 3-letter word appears twice? _____ What English word could it be? _____

b If you are right, what shift turns T into B? _____

c Decode the first word of the message with that shift: _____

No.5

Two shifts in a row. Shifts add up like jumps on a 26-hour clock.

a Shift 6, then shift 9 is the same as one shift of _____ . (b) Shift 19, then shift 12: one shift of _____ .

c Ben codes a message with shift 9, then codes the result with shift 17. What does he get? Why?

.....

d Which shift (not 0), used twice, always gives back the original message? _____

FINISH PAGE

Name _____ Group _____ Date _____

Everyone gets to this page. The investigation is for anyone who wants it, at any time. The exit slip and looking back are for everyone.

Investigation **Is it always true?**

Code the word CAT with a shift of 8. Then code your answer with a shift of 18. What do you get? Try two other shifts that add up to 26. (a) Is it always true that two such shifts undo each other? Explain. (b) Is there a shift that is its own decoder, so that coding twice gives back the original? Find ALL of them.

First step: code the letter C with shift 8. Then shift your answer by 18. Where do you land?

Exit slip **Code the word HI with a shift of 25. What do you notice?**

Looking back About 2 minutes. Nobody marks this. You can write "pass" for any line.

Something I did well _____

The hardest part was _____

What helped me: [ ] I drew or built it [ ] I tried a smaller example [ ] I checked my answer [ ] I asked for help

[ ] I am still stuck (my tutor will check in during the lesson) [ ] pass

STORY CORNER **Al-Kindi** (Iraq, about 801 to about 873)

Al-Kindi worked in Baghdad about 1,200 years ago. Around the year 850 he wrote the earliest explanation we know of for breaking a code by counting letters, because some letters are used more often than others. His manuscript was lost for a long time before a modern scholar found it in Istanbul.

Try it (optional): Every letter was shifted by the same amount. The commonest letter in the code is L. If it stands for E, what was the shift? (A = 0 … Z = 25, mod 26)

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Choose any question. You can start here after question 2, or when you finish the Start here pages. You can go back to Start here at any time. Everyone joins Talk time and the exit slip.

No.1

The pattern CODECODECODE... goes on and on. What is the 100th letter?

No.2

It is 10 o'clock now. What time will it be 1000 hours from now (on a 12-hour clock)?

No.3

Find the smallest number bigger than 10 that leaves a remainder of 3 when it is divided by 4 and also when it is divided by 5.

No.4

(a) Work out 29 mod 12, 50 mod 7 and 100 mod 26. (b) Decode the whole message from Q4 (BPM SMG QA CVLMZ BPM UIB) with the shift you found.

Go further

No.5

Write your own secret message of at least 8 letters and code it with a shift of your choice. Write clear instructions telling a friend how to decode it, and explain how a friend could crack it without being told the shift. Swap and decode.

No.6

Ali codes a message with a shift of 7. Mei codes Ali's result with another shift. The final message is the same as the original coded with a shift of 2. What shift did Mei use?

No.7

Start at 12 on a clock and keep jumping forward 5 hours. Do you land on every number before you get back to 12? What about jumps of 4 hours?

DATA DETECTIVES

Start here Know this idea already? You may start with Think harder. Or after question 4 choose: carry on, or try Think harder.

Name _____ Group _____ Date _____

No.1

Five friends counted the books they read in the holidays: 4, 7, 2, 8, 9

a What is the total? _____ (b) What is the mean? _____

c Balance check: how far is each value above (+) or below (–) the mean?

4 → _____ 7 → _____ 2 → _____ 8 → _____ 9 → _____ Add them up: _____

No.2

Siti played 4 games. Her mean score was 17. Her first three scores were 15, 20 and 14.

a Her total for the 4 games? _____ (b) Her 4th score? _____

c She plays a 5th game. What must she score to make her mean exactly 18? _____

No.3

The data 6, 8, 10, 12, 14 has a mean of 10. Predict first: up, down or the same? Then work out the new mean.

Change	Up, down or the same?	New mean
(a) Add a new value, 10.		
(b) Add 3 to every value.		
(c) Remove the 14.		
(d) Change the 14 to 44.		

No.4

Seven P6 pupils get this pocket money each week: $10, $12, $12, $15, $16, $20, $69

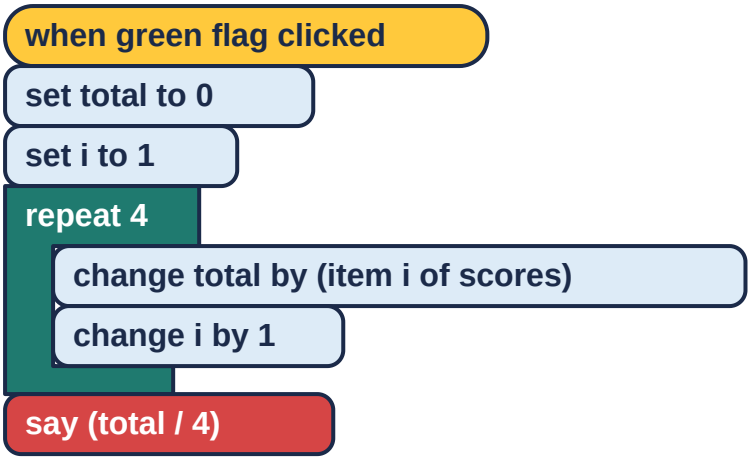
a Mean (total $154) _____ (b) Median _____ (c) Mode _____

d How many of the pupils get less than the mean? _____

e Jun says, "A typical pupil gets $22 a week." Is that fair? Which average describes a typical pupil better? Explain.

No.5

Trace the code. The list called scores holds 5, 9, 6 and 12 (item 1 is 5, item 2 is 9, and so on). Fill in the trace table, one row for each time round the loop.



Loop	i	item i	total
1st			
2nd			
3rd			
4th			

a What does the sprite say? _____

b Jo adds a 5th score, 13, to the list. Which two blocks must change so the program works for a list of ANY length? What should it say now?

.....

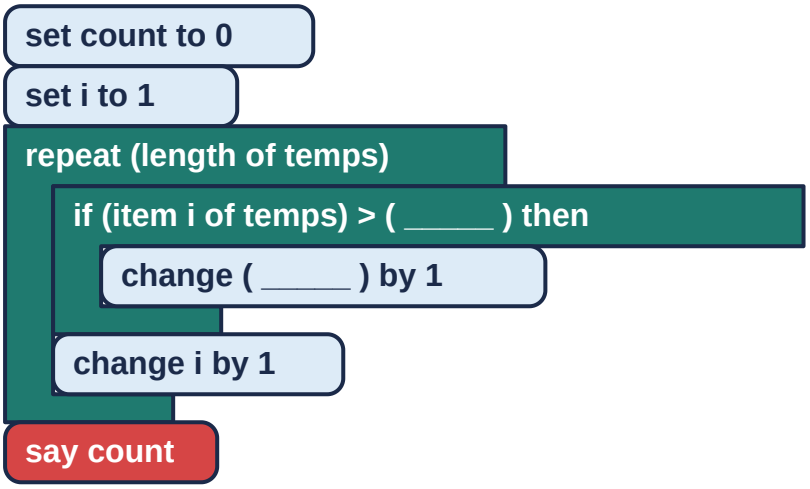
.....

No.6

The highest temperature each day for a week in Singapore, in °C: 29, 31, 33, 30, 32, 34, 28

a Mean temperature? _____ °C (b) Days hotter than the mean? _____

c Plan your code. The variable mean already holds your answer to (a). Fill in the two blanks so the program counts the days hotter than the mean. What will it say?



Optional, for children who already code. Python (optional): put the temperatures in data. Does it agree with (a)?

```
data = [3, 0, 4, 1, 2]
total = 0
for value in data:
    total = total + value
mean = total / len(data)
print("Mean:", mean)
```

Blank 1: _____ Blank 2: _____ It says: _____

FINISH PAGE

Name _____ Group _____ Date _____

Everyone gets to this page. The investigation is for anyone who wants it, at any time. The exit slip and looking back are for everyone.

Investigation **What if one number changes?**

Data: 4, 6, 7, 9, 14. Find the mean, the median and the mode (is there one?). Now change the 14 to 24, then change it to 4, then to 7. Make a table of the mean, median and mode each time. What changes? What stays the same? Can you change TWO numbers and keep the mean the same?

First step: find the mean of 4, 6, 7, 9 and 14 (add them up, then share equally).

Exit slip **The mean of 3 numbers is 9. Two of them are 5 and 12. What is the third number? A fourth number, 13, is added. What is the new mean?**

Looking back About 2 minutes. Nobody marks this. You can write "pass" for any line.

Something I did well _____

The hardest part was _____

What helped me: [ ] I drew or built it [ ] I tried a smaller example [ ] I checked my answer [ ] I asked for help

[ ] I am still stuck (my tutor will check in during the lesson) [ ] pass

STORY CORNER **Adolphe Quetelet** (Belgium, 1796 to 1874)

Adolphe Quetelet began teaching maths at seventeen and later directed the new observatory in Brussels. In 1835 he wrote about the average man, an idea made from many measurements and not a real person. In 1853 he helped organise the first International Statistical Congress.

Try it (optional): Heights 138, 142, 145, 150 and 155 cm. Find the mean. A sixth child of 164 cm joins: what is the new mean?

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Choose any question. You can start here after question 4, or when you finish the Start here pages. You can go back to Start here at any time. Everyone joins Talk time and the exit slip.

No.1

The mean height of 4 children is 150 cm. A fifth child joins the group and the mean height becomes 152 cm. How tall is the fifth child?

No.2

Five whole numbers have a mean of 8, a median of 10 and a mode of 12 (12 appears more often than any other number). Find them. Is there more than one answer?

No.3

Write 5 whole numbers with a mean of 6 and a mode of 4.

No.4

A Scratch list holds 8 numbers with a mean of 12. Then the number 21 is added to the list. What will the mean program say now?

Go further

No.5

Make up your own “mean mystery” like the Think harder mean puzzles: choose five whole numbers, then write clues (any of mean, median, mode, smallest, largest) that give ONLY your numbers. Swap with a partner. Can they find a different set that fits your clues?

No.6

The mean of 1, 2, 3, ... , 99 is 50. Explain why without adding them all up. Then plan a short program that checks it.

No.7

A group of 8 children has a mean mass of 41 kg. When their tutor joins, the mean mass of all 9 becomes 44 kg. Then one more person joins and the mean of all 10 becomes 43 kg. What is the tutor’s mass? What is the mass of the last person to join?

SHOWCASE CHALLENGE

Start here Know this idea already? You may start with Think harder. Or after question 3 choose: carry on, or try Think harder.

Name _____ Group _____ Date _____

No.1

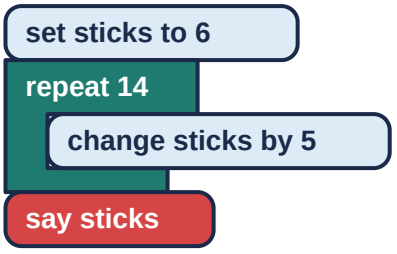
Chains of hexagons are made from sticks. Start with (a): the picture shows 1, 2 and 3 hexagons.



1 hexagon
6 sticks

2 hexagons
11 sticks

3 hexagons
16 sticks



a Sticks for 4 hexagons? _____ (count on from the picture) 10 hexagons? _____

b Rule for n hexagons: sticks = _____

c How many hexagons can be made with exactly 101 sticks? _____

d The code says _____ . It checks a chain of _____ hexagons.

No.2

Prime hunters.

a I have 2 digits and exactly 3 factors. My digits add up to 13. What am I? _____

b Write 84 as a product of prime numbers. _____

c Find ALL the ways to write 40 as two primes added. _____

No.3

Quick-fire round.

a It is –4 °C. It warms up 9 °C, then cools 12 °C. What is it now? _____

b A machine does × 3, then – 10. Which IN number comes OUT unchanged? _____

c A turtle draws a regular polygon, turning 24° at each corner. Sides? _____

d It is 9 o'clock. What time will a 12-hour clock show 100 hours from now? _____

No.4

Maths first! Work out what each starter project should do BEFORE you run it.

a Mystery Machine (× 3, then + 4). n = 9 gives? _____ OUT 58 came from? _____

b Prime or Not? How many factors does it count for 91? _____ Prime? _____

c Lucky Seven, changed to count 8s: about how many 8s in 1000 rolls? Why?

.....

No.5

My puzzle program: the plan.

My idea (Starter A, B, C or my own)	
The maths puzzle it asks	
Its answer, worked out by hand	
My twist (what I change or add)	

No.6

Plan your blocks (under 15) in the box. Fill in each test BEFORE you run it.

Python fans (optional): Starter A. To test it with Q4(a), change line 2 to n = 9

```
import random
n = random.randint(1, 20)
out = n * 3 + 4
print("My machine does x 3, then + 4.")
print("Out came", out)
guess = int(input("What went in? "))
if guess == n:
    print("You cracked it!")
else:
    print("No, it was", n)
```

Input (what I type or set)	It should say (by hand)	It said	Same? ✓
Test 1			
Test 2			
Edge case			

Ready to show? [ ] I can explain the maths [ ] I tested a value I knew [ ] I tested an edge case

FINISH PAGE

Name _____ Group _____ Date _____

Everyone gets to this page. The investigation is for anyone who wants it, at any time. The exit slip and looking back are for everyone.

Investigation **Primes in a sequence**

The sequence 4, 11, 18, 25, ... has the rule 7n – 3. Find ALL the prime numbers in this sequence up to 100. Show your list of terms and how you tested each one. Then explain why no term of the sequence is a multiple of 7.

First step: write the first 5 terms of the sequence 7n – 3 (4, 11, 18, ...).

Exit slip **Write one thing your program does, one test you did (the input and the answer you expected), and the maths that explains it.**

Looking back About 2 minutes. Nobody marks this. You can write "pass" for any line.

Something I did well _____

The hardest part was _____

What helped me: [ ] I drew or built it [ ] I tried a smaller example [ ] I checked my answer

[ ] I asked for help

[ ] I am still stuck (my tutor will check in during the lesson) [ ] pass

STORY CORNER **Ada Lovelace** (United Kingdom, 1815 to 1852)

Ada Lovelace translated an 1843 article about Charles Babbage's Analytical Engine, a machine that was never finished, and added notes of her own. One note shows step by step how it could work out Bernoulli numbers. This is often called the first computer program, but historians debate that name.

Try it (optional): Trace like a computer: n = 2. Repeat 3 times: change n to n x 2 + 1. What is n at the end?

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Choose any question. You can start here after question 3, or when you finish the Start here pages. You can go back to Start here at any time. Everyone joins Talk time and the exit slip.

No.1

A row of squares uses 4, 7, 10, 13, ... matchsticks. Which pattern number uses exactly 100 matchsticks?

No.2

Find all the prime numbers between 80 and 100.

No.3

A regular polygon has an angle of 150° inside each corner. How many sides does it have?

No.4

Cards 1, 2, 3, 4 make 2-digit numbers (no card twice). (a) How many numbers? How many are even? (b) One of the numbers is picked at random. What is the probability it is a multiple of 3?

Go further

No.5

Make up a challenge question that mixes TWO topics from the programme (for example primes and the clock, or sequences and negative numbers). Write the question, the answer and the working. Swap with a partner and solve each other's.

No.6

A spinner has the numbers 1 to 8, all equally likely. Ravi spins it twice and adds the numbers. What is the probability that the total is 9? Plan a program that tests it 1000 times.

No.7

The nth term of a sequence is 6n – 1. Is 2027 (the year you start Secondary 1) in the sequence? Which is the first term bigger than 2027?

Cut-outs

For the Level 6 program. Cut along the dashed lines and keep the pieces in one envelope.

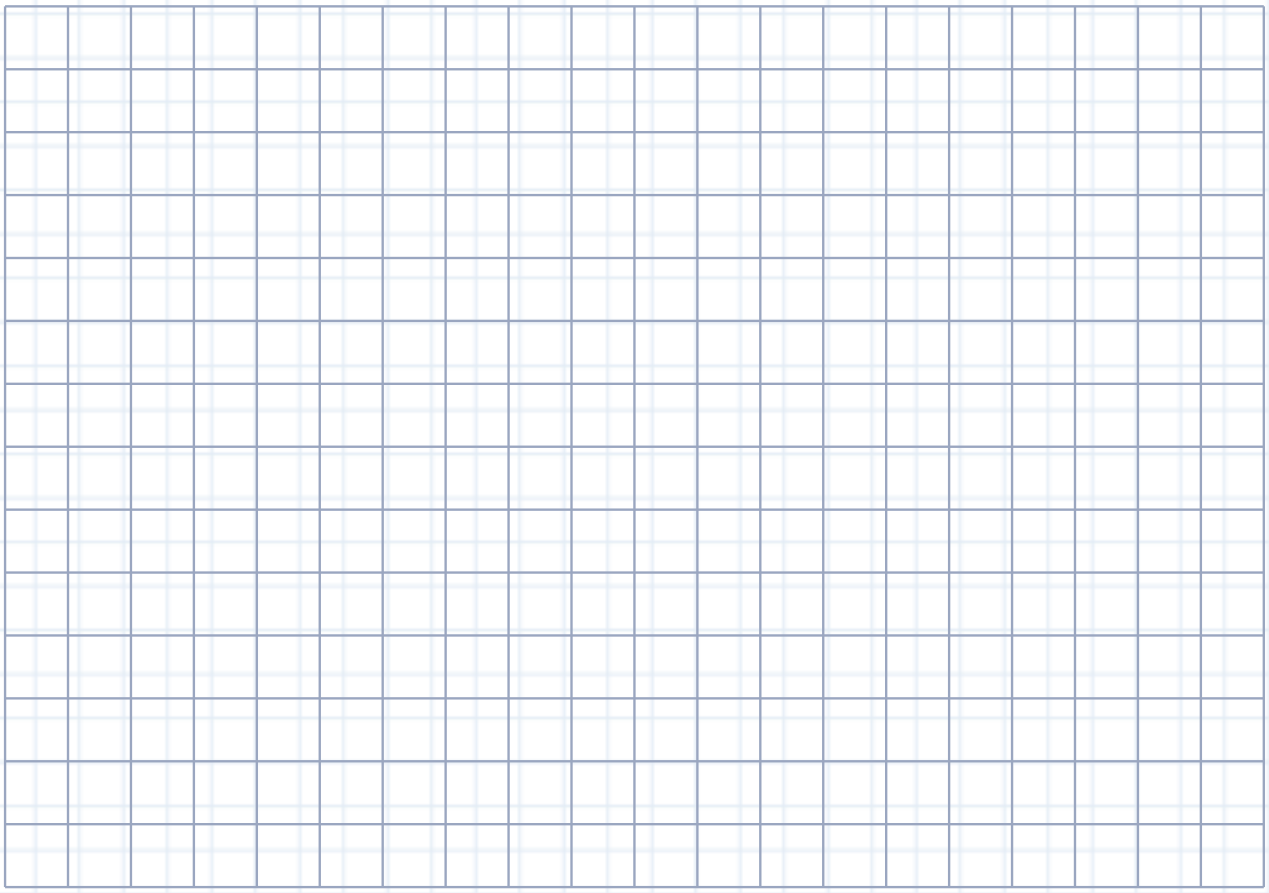
Unplugged code cards (Lessons 1, 4 and 7)



Unplugged coding: cut out the blocks and snap them into a program on the table before you build it on scre

Square grid

For Lessons 7 and 9.



CERTIFICATE OF COMPLETION

Math Inquirers Holiday Program

Level 6 Holiday Program

This certifies that

.....

completed all 10 lessons, solved puzzles, spotted patterns and explained their thinking.

Notice it. Test it. Explain why.

Tutor: _____ Date: _____